Sensing GUI

60 GHz radar system platform

About this document

Scope and purpose

This application note describes Sensing GUI, a Graphical User Interface (GUI) of Infineon's 60 GHz radar system platform. After a quick start guide, the document explains the different modules and features of the software.

Intended audience

This document is intended for anyone working with the Sensing GUI in Infineon's 60 GHz radar system platform.

Table of contents

1 Quick start guide

Sensing GUI is a Graphical User Interface (GUI) of Infineon's 60 GHz radar platform. It enables configuration of the BGT60TRXx sensors.

After connecting a board that is supported by the 60 GHz radar platform (RS, 2019) via USB to the host PC, the board should appear as a COM port in the Windows operating system. On old Windows versions, it might be necessary to install the board's driver manually. This driver is supplied with the board. By using the "Device Manager" in Windows, it is possible to verify that the board's driver is correctly installed.

Sensing GUI does not need to be installed but can be launched directly from the directory on the hard drive. After starting up the GUI, one can find the available COM ports in the menu bar under "Connection". An example is shown in Figure 1.

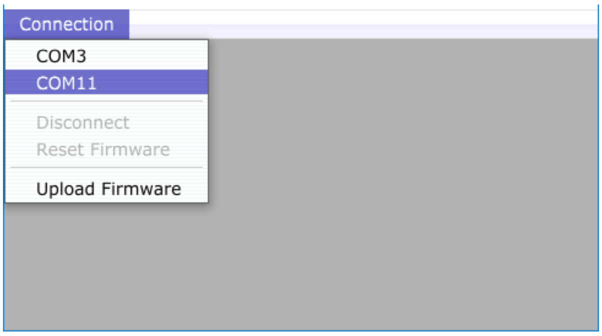

Figure 1 Sensing GUI's start-up screen

After selecting the COM port of the board, the main screen of Sensing GUI will appear, shown in Figure 2. To execute the standard configuration, simply click the "Trigger" button on the top-left side of the GUI.

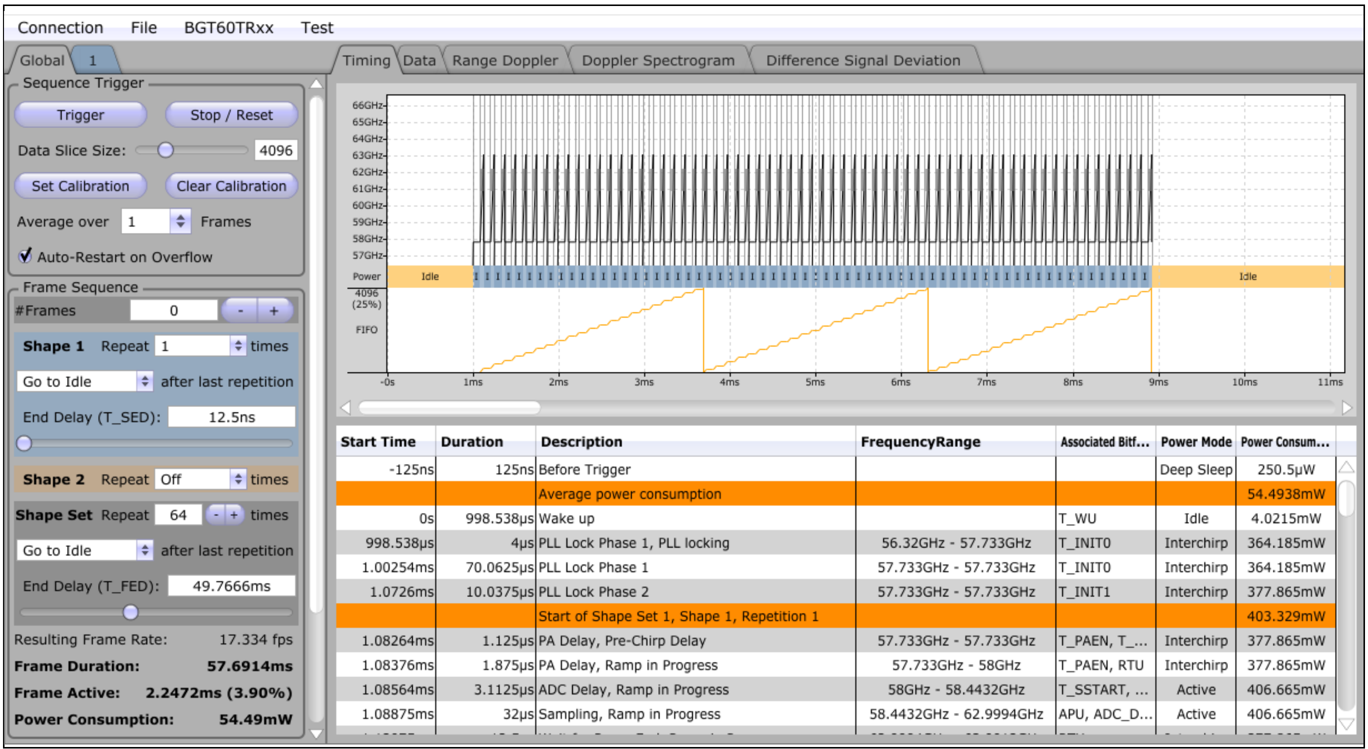

Figure 2 Sensing GUI's main screen

1.1 Data tab

In the "Data" tab, the time domain data of the acquired radar data is displayed as well as the frequency domain data – see Figure 3 for details. In the top part of the tab, the time domain data is displayed. To zoom in and out of this view, put the cursor above it and use the mouse wheel. As described in section 2.1, this view is used to verify that there is no clipping with a large signal swing to use most of the available ADC range.

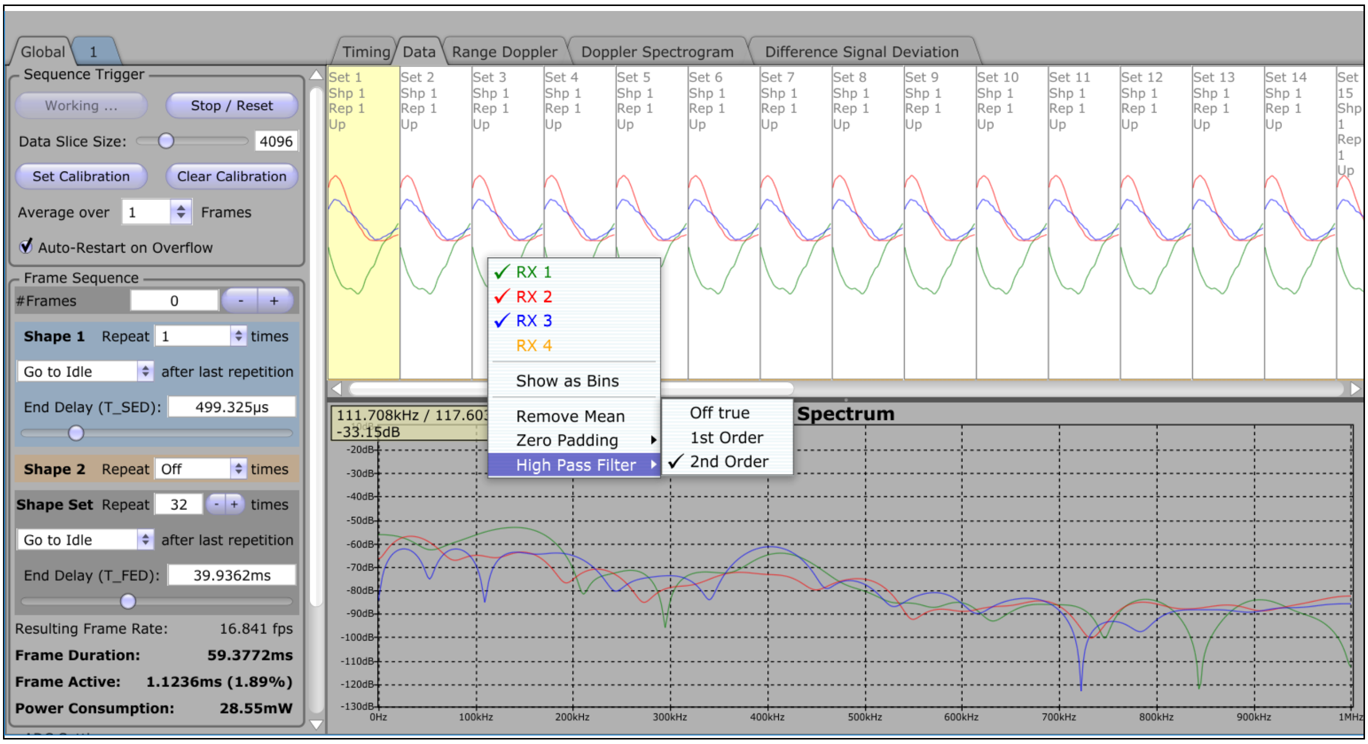

Figure 3 **The "Data" tab shows the time signal in the top half and its spectrum in the bottom half**

In the bottom part of the tab, the frequency domain data is displayed. By right-clicking in the graph, you can configure the displayed spectrum. Then, you can choose the displayed RX-channels, the type of plot, whether to remove the mean, the zero padding length and what kind of high-pass filter to apply.

1.2 Range-Doppler tab

To work with the "Range Doppler", first you need to configure it. Right-click in one of the plots and select "Source" – "Chirp 1 of each set", as shown in Figure 4.

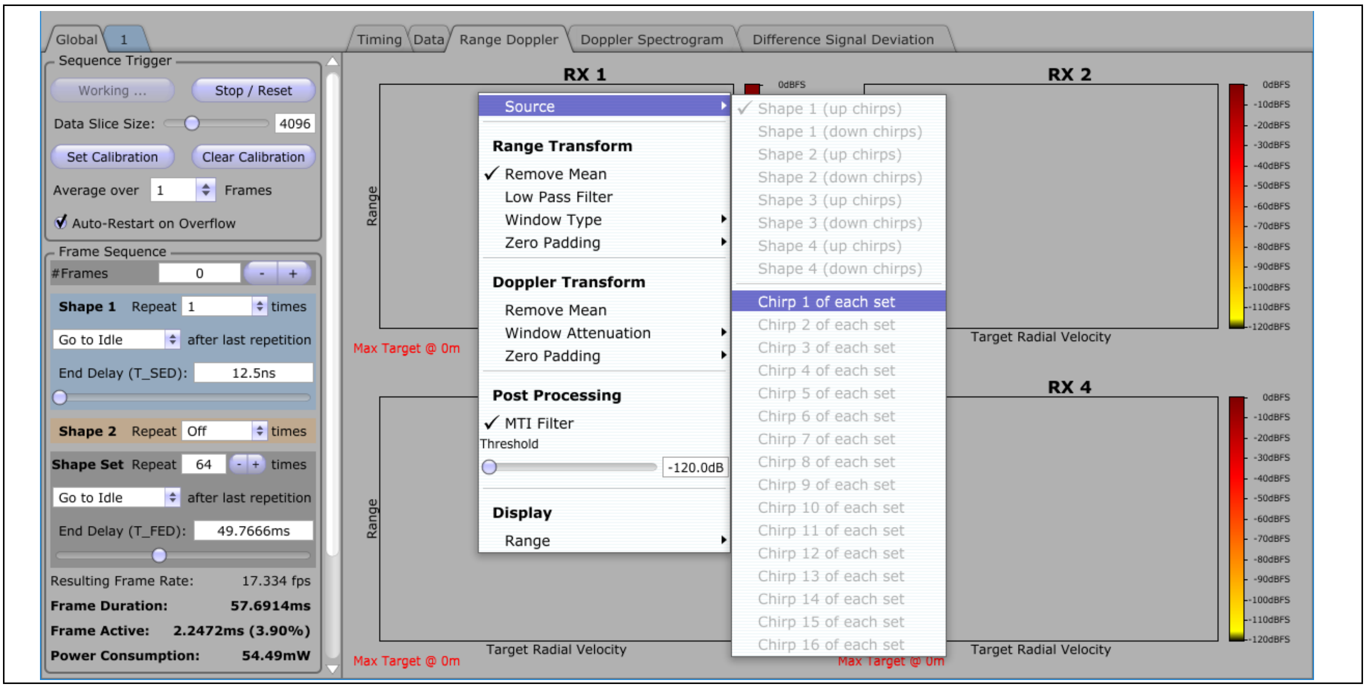

Figure 4 Configuring the source "Range Doppler" tab

To remove data from the noise floor, you can adjust the noise floor by right-clicking in one of the plots and selecting "Threshold" and then the desired value.

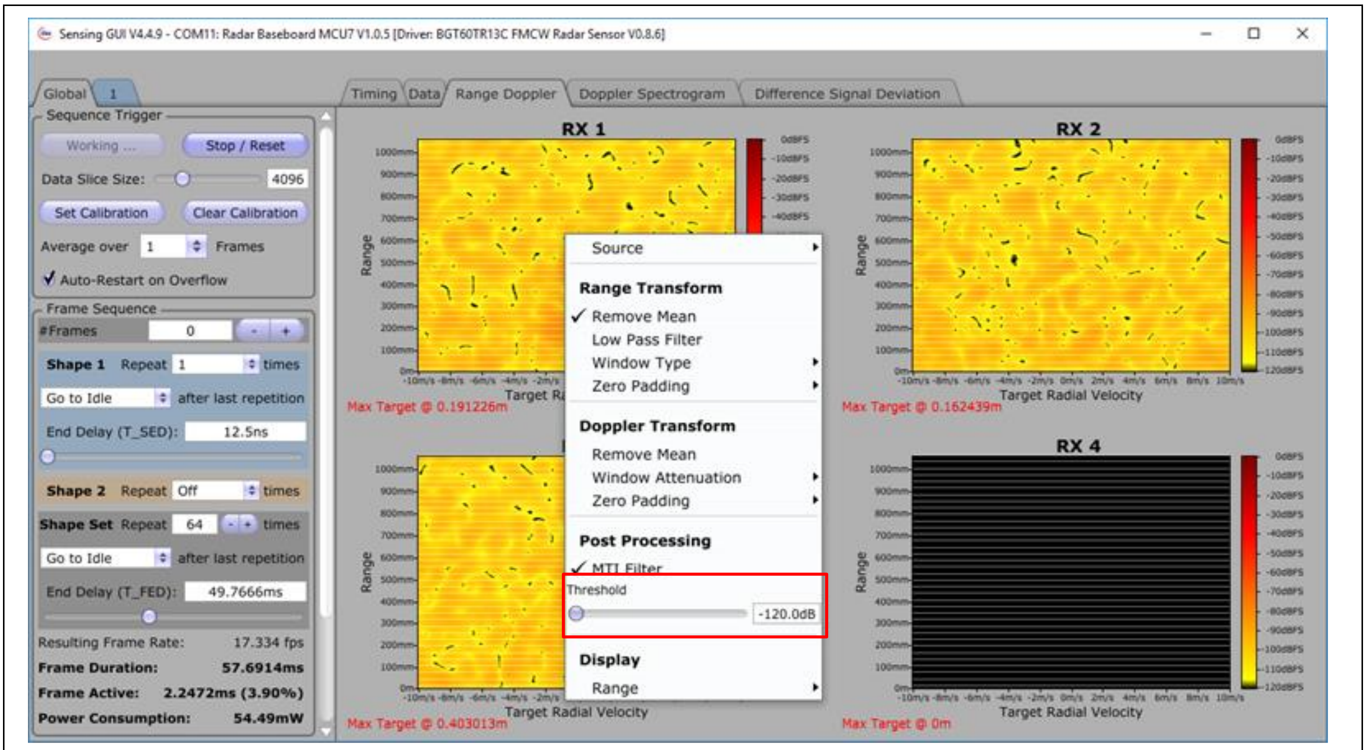

Figure 5 Configuring the threshold of the Range Doppler maps of the acquired data

The "Range Doppler" tab will display the Range Doppler maps of all activated antennas. The color map gives an indication of the target strength. The vertical axis is the radial distance between the radar sensor and the target, and the horizontal axis is the radial velocity of the target, meaning the velocity components directly toward the radar sensor. A feature of this view is that it contains a filter that removes static targets. This static target removal helps with data visualization. However, this means it also removes desired targets that are not moving.

An example of a Range Doppler map is shown in Figure 6. There, two targets are visible above the threshold. Both have a similar strength of ~60 dBFS and a similar radial distance from the radar of ~300 mm. However, their radial velocities are different. One is moving at ~0.6 m/s away from the radar sensor, whereas the other one is approaching at ~0.6 m/s.

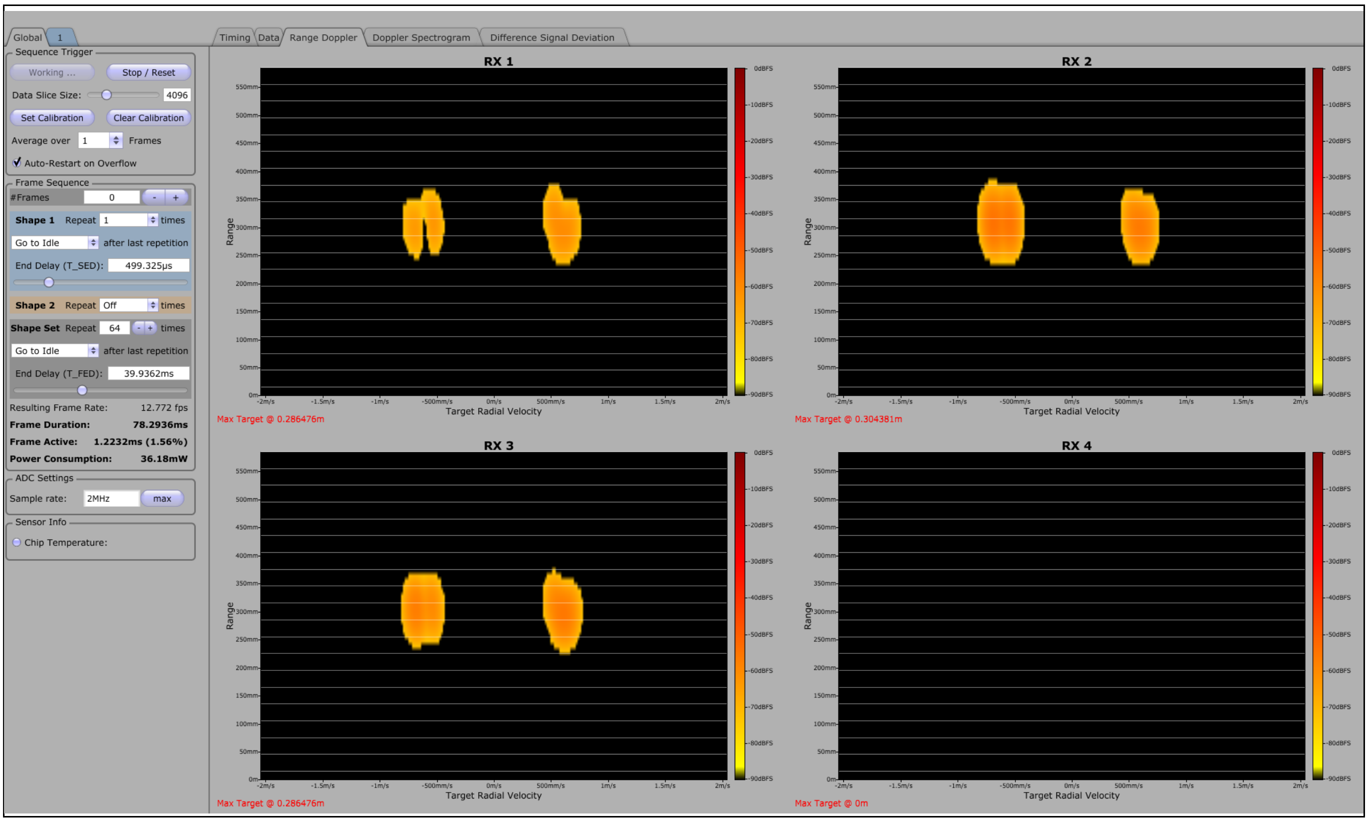

Figure 6 **Example Range Doppler map with two targets at roughly the same distance of 300 mm but at different radial velocities, ±0.75 m/s**

Another parameter to configure in this view is the “Doppler Attenuation”. Typically, the slow-time FFT also causes side lobes. The Doppler attenuation gives an indication of how strongly these side lobes are suppressed. An example of a Range Doppler map with low Doppler attenuation is shown in Figure 7.

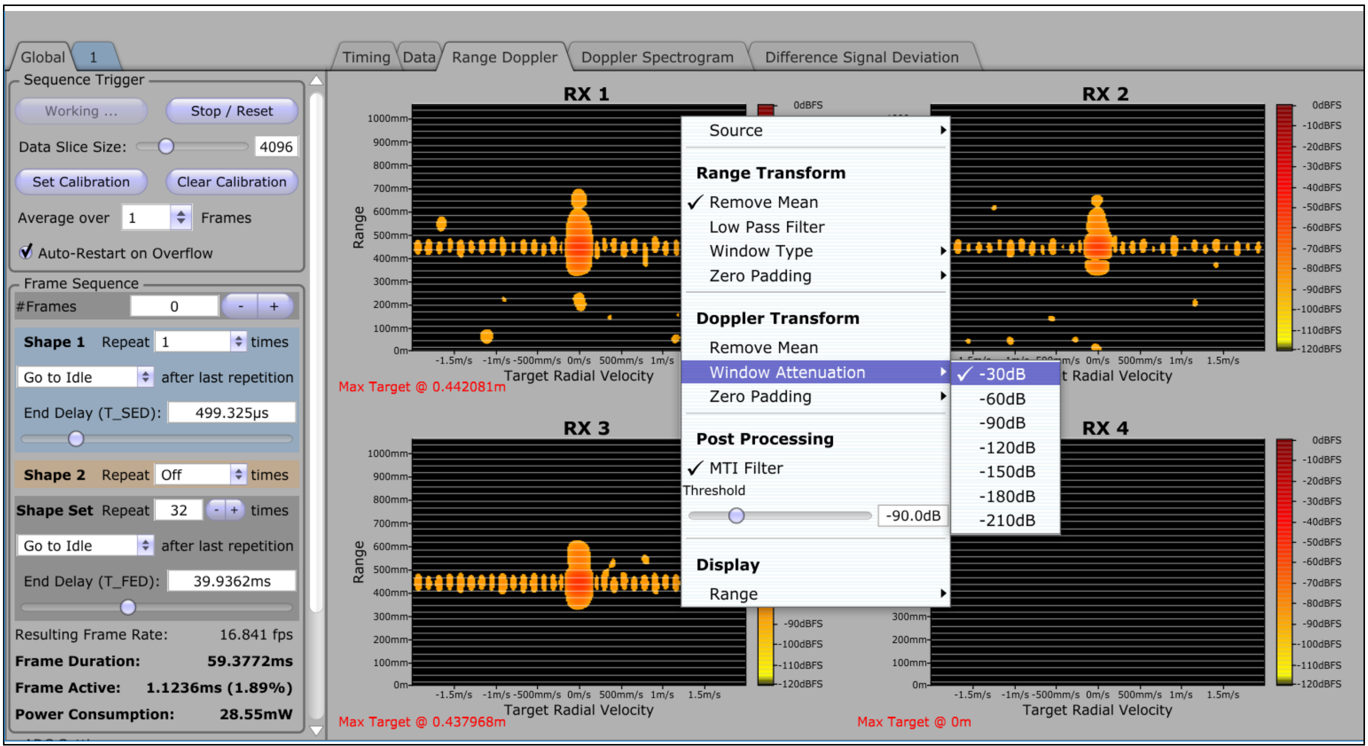

Figure 7 **Range Doppler map with low Doppler attenuation**

This menu also lets the user configure other settings too like choosing different window function, enabling the Low Pass Filter and so on. Zero padding factor can also be changed to improve the accuracy of radar. Increasing the zero padding would make the FFT bin size smaller, leading to better accuracy. For most applications, the sweet spot between computational effort and displayed accuracy is at a 4th multiple of the 'Samples per chirp'. Furthermore, the Moving Target Indicator (MTI) filter can be enabled to detect only the moving targets and ignore any stationary targets.

1.3 Doppler Spectrogram tab

Another option to display radar information is with a spectrogram in the "Doppler Spectrogram" tab. There, the strongest detected target is taken and its radial velocity is plotted over time. In the "Doppler Spectrogram" tab, you must configure the source and threshold again by right-clicking in one of the plots and selecting the source and threshold (Figure 8). An example of a Doppler Spectrogram of a target that alternates between approaching and receding from the radar sensor is shown in Figure 9.

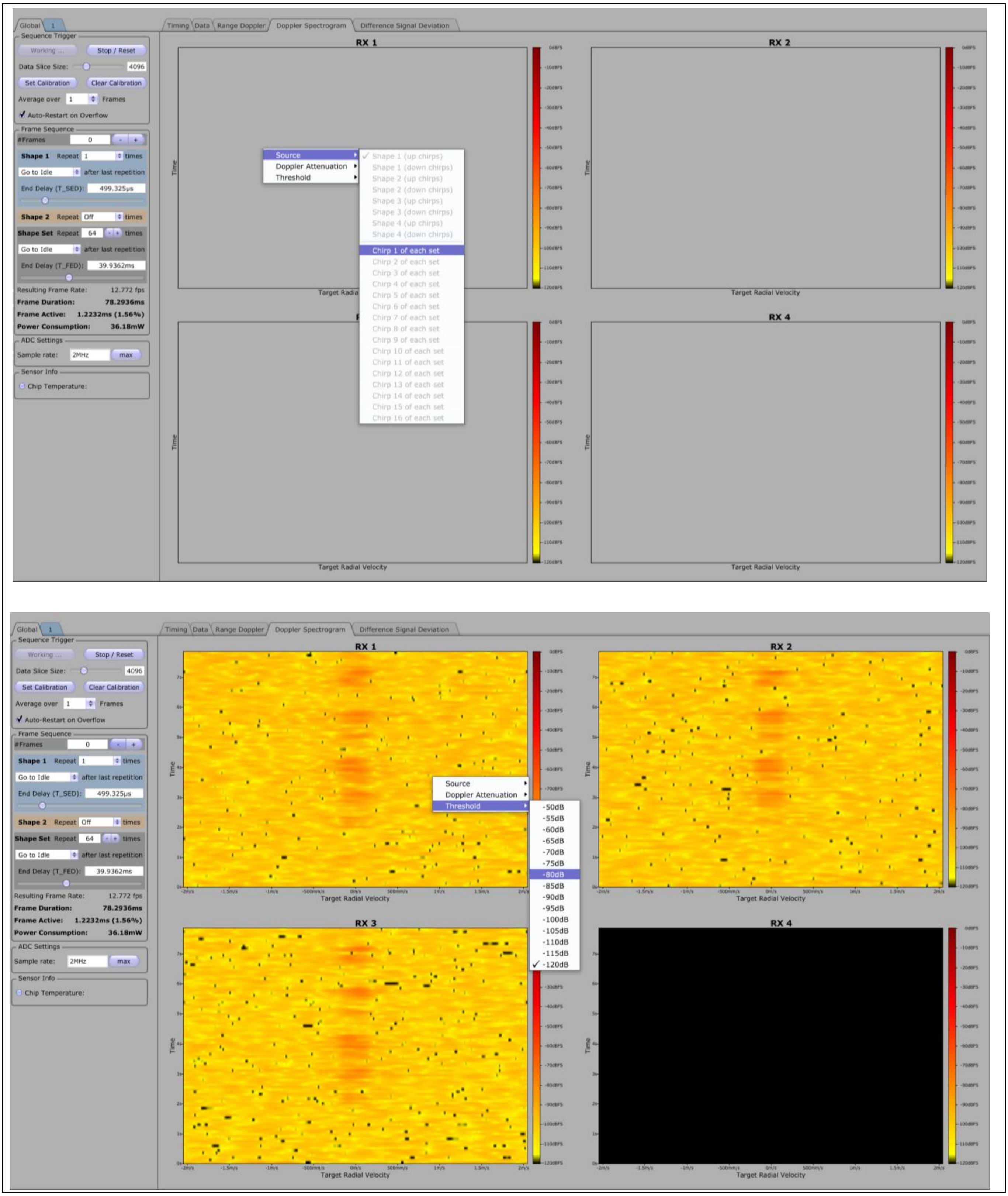

Figure 8 **Configuring the source and threshold "Doppler Spectrogram" tab**

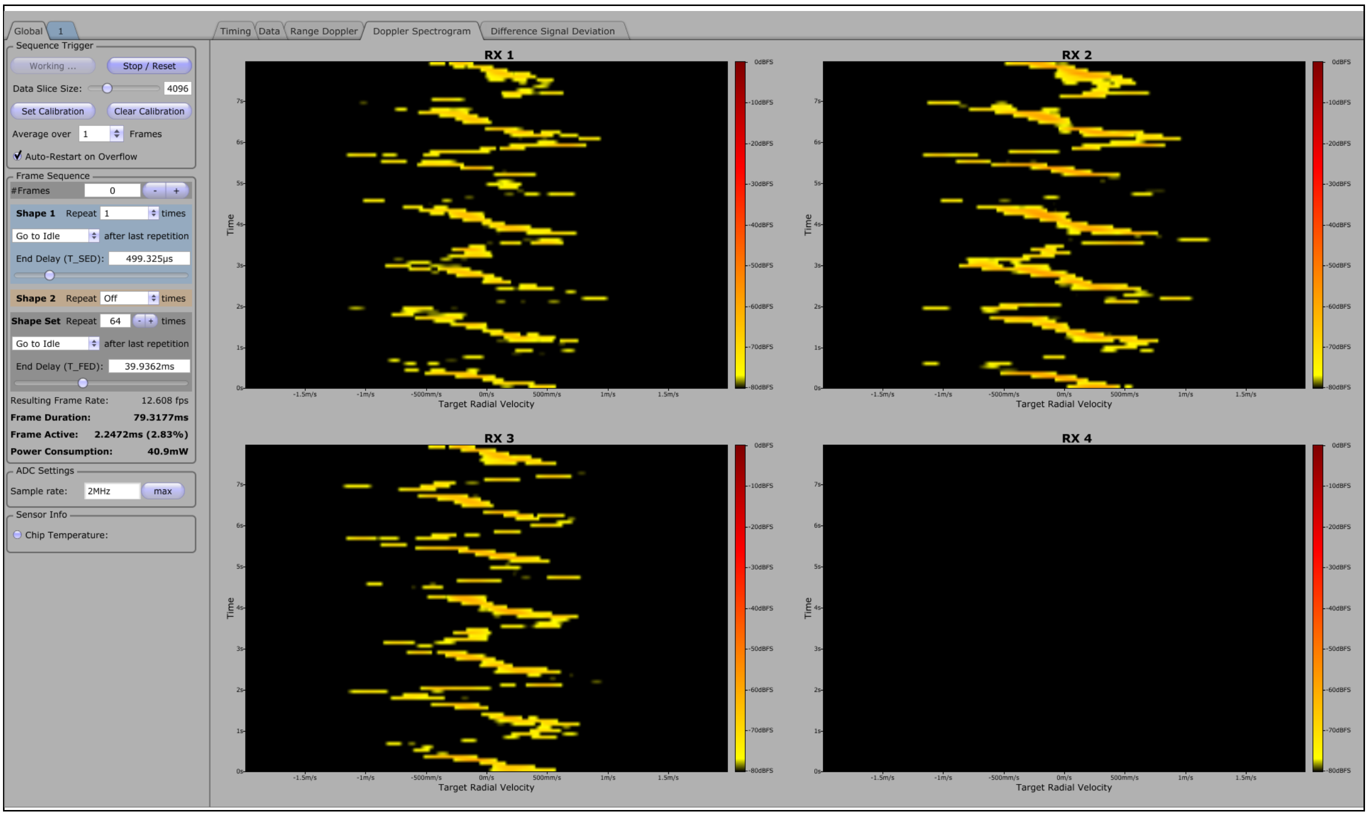

Figure 9 A Doppler Spectrogram of a target that alternates between approaching and receding

2 Chip configuration

Figure 2 shows that located on the left side of Sensing GUI's main screen is the chip configuration. On the right side is the data visualization. In Figure 2, you can see the "Timing" tab, which shows the timing of the set chip configuration. The effect of a change in the chip configuration will be displayed there.

2.1 Shape configuration

The configuration of BGT60TRXx is divided into shapes. The chips support up to four different shapes that can be repeated. For details about shape set configuration, see section 2.2.

Each activated shape will appear as a tab in the shape selector, displayed in Figure 10. When selecting a shape tab, the corresponding shape can be configured. First, choose a chirp direction. The BGT60TRXx supports four different directions: "up", "up-down", "down-up" and "down". The different directions and their resulting timings are displayed in the timing tab.

The bandwidth of a chirp can be configured with a text edit field or two sliders for the minimum and maximum frequency, as illustrated in the bandwidth selector of Figure 10. The TX-power can be adjusted with the TX-power slider. This enables a coarse tuning of the TX-power, with 31 being the maximum power and 0 being the minimum power.[1] Each direction of each shape (e.g. "up"-direction of an "up-down" shape) allows the configuration of the acquired number of samples as well as the antenna configuration used for the direction of the shape, as shown in Figure 10.

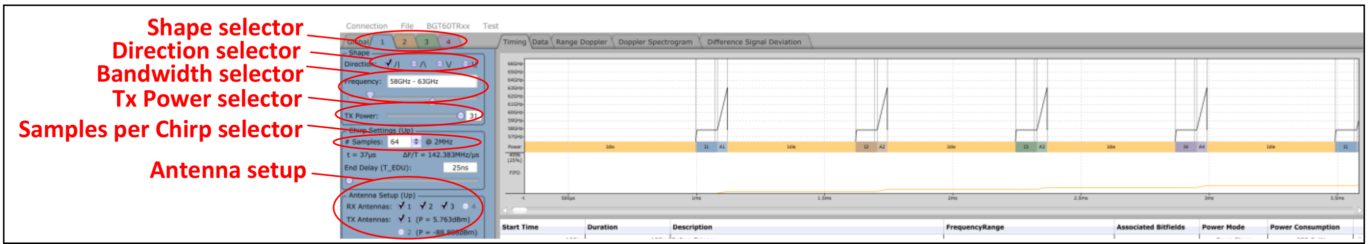

Figure 10 Shape configuration in standard mode

To switch to the expert mode of Sensing GUI, you must select "BGT60TRXx" in the menu bar and activate "Show Expert Parameters", as shown in Figure 11. In this new view, it is possible to adjust the intermediate frequency (IF) gain and the IF cut-off frequency for each direction of a shape. The high-pass cut-off frequency can be configured between 20 kHz and 80 kHz. The IF amplification is done in two stages. For better noise performance, it is recommended to increase the noise of the first stage (HP gain) before fine-tuning the gain with the second stage (VGA gain). After bandwidth, TX-power, number of samples and antenna configuration have been chosen, adjust the gain of this configuration. Therefore, activate the "Data" tab and adjust the IF settings such that the signals of all antennas are maximized without clipping. Figure 11 shows an example, where shape 1 is optimized. Shape 2 has too-low amplification, and shape 3 has clipping because of the excess amplification.

[1] This tuning is highly non-linear.

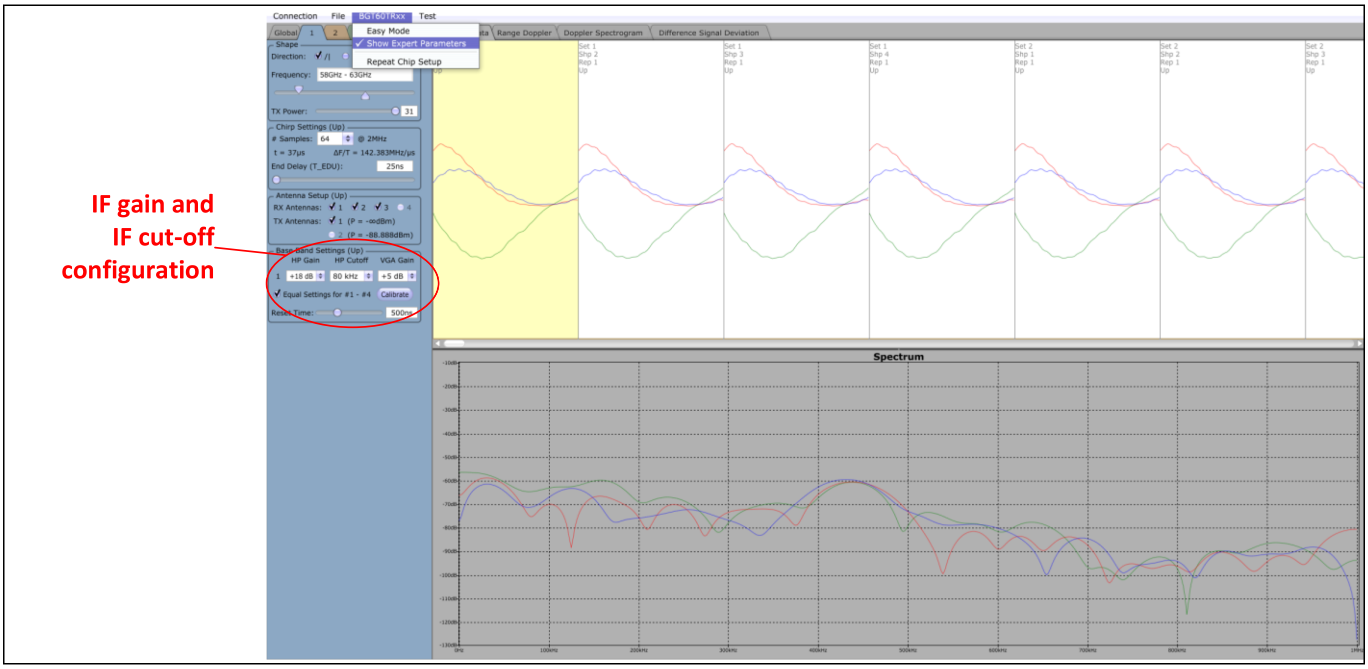

Figure 11 Shape configuration in expert mode

2.2 Shape set configuration

The shape set configuration is part of the “Global” tab, shown in Figure 12. The effects of the changes are illustrated in the “Timing” tab. To highlight the features of the shape set configuration, an example consisting of four shapes is configured, shown in Figure 12. Initially, the chip is in deep sleep mode with a power consumption of much less than 1 mW. When booting up the chip, some self-configuration must be executed, and this takes nearly 1 ms (998.538 µs, to be precise). After this, the PLL must be locked before a chirp can be run. This locking procedure requires about 70 µs.

The first shape is configured to be repeated twice. The repetition is performed without pauses between the chirps. After the repetition, the chip is configured to go to idle mode for a period of 499.325 µs (end delay). The idle mode is a power-saving mode that disables the whole chip except the digital domain and the bandgap for MADC. The power consumption in this mode about 5 mW. This ends shape 1, and the chip continues with shape 2.

Shape 2 is executed only once. Before this execution, the PLL must be relocked. This again takes about 70 µs. After running the shape configuration, the chip stays active for 499.324 µs (end delay). In active mode, the High-Frequency (HF) part of the chip is not disabled. The power consumption in this mode is about 300 mW.

Given that the HF part of the chip stayed active at the end of shape 2, shape 3 can be executed immediately. This execution is configured to be repeated once and then the chip enters deep sleep mode. It stays in deep sleep mode for 499.325 µs (end delay). When returning from deep sleep, the chip must run self-configuration again. This takes about 1 ms, as mentioned above.

After the self-configuration, the chip has to relock the PLL (~85 µs) before shape 4 can be executed. After shape 4, the chip enters idle mode for 499.325 µs. As the shape set is configured to be repeated, all shape sets are configured to be repeated twice. The difference between the first and second execution is that in the example of Figure 12, the execution starts from idle mode and not deep sleep (no reconfiguration required). At the end of the frame (after the second execution of shape 4), the mode and time after shape 4 is ignored. This chip uses the mode and time configured under shape set (meaning an end-of-frame delay). In this example, that is deep sleep mode for 1.99695 ms.

Below the configuration in Figure 12, you can see frame parameters such as frame repetition rate, frame duration, chip active time during frame, or the estimated power consumption of this configuration.

Typical configurations (Figure 2) are not so complicated and contain only one shape that is repeated with a certain spacing (setting the maximum velocity in Range Doppler) for a certain number of times (setting the velocity resolution in Range Doppler) with a specific frame end delay (to set the frame repetition rate).

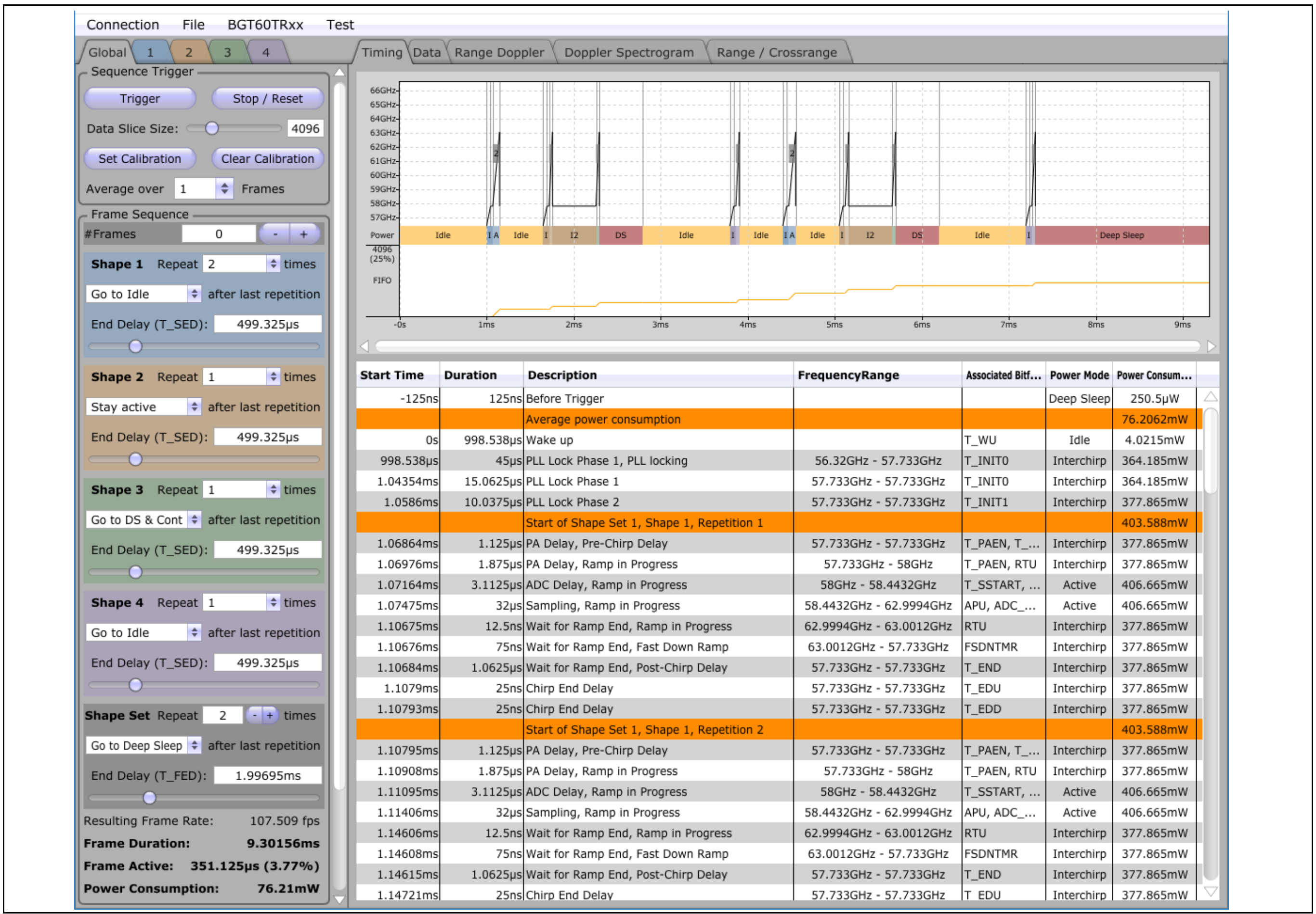

Figure 12 Shape set configuration

2.3 Auto restart on overflow

When configuring frames with very short pauses in between shapes and at the end of the frame, the data to transfer from the board to the host PC can exceed the maximum data rate for USB devices when using a USB 1.1 device or the PC's operating system maximum data rate for serial communication when using a USB 2.0 device. Sensing GUI will reacts by showing a **FIFO memory overflow**. The GUI has an option to restart the acquisition of chirps automatically after a FIFO memory overflow is detected as shown in Figure 13. This can be enabled or disabled using the 'auto restart on overflow' feature on the 'Global' tab.

Note: *There could be a slight delay between the chirps during this 'restart' which could lead to a longer interval between the chirps. In the scenario where strictly equidistant chirps are needed, it is recommended to disable this feature. To solve this issue without the auto restart feature, data rate must be decreased. The simplest way to reduce the data rate is to increase the frame end delay (T_FED) in the shape set configuration.*

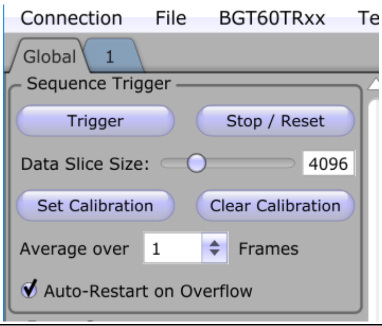

Figure 13 Auto restart on overflow

2.4 ADC configuration

In standard view (Figure 14), it is very simple to adjust the ADC sample rate by typing the desired value into the edit box. Consider that the BGT60TRXxC has an anti-aliasing filter of 500 kHz. Hence, all signals above 500 kHz are attenuated.

Figure 14 ADC configuration

2.5 CW mode operation

To put the radar sensor in CW mode operation, you must switch to expert mode by selecting “BGT60TRXx” in the menu bar and activate “Show Expert Parameters”, as shown in Figure 11. At the bottom of this view, you can find the “test mode” for CW mode operation, shown in Figure 15. There, you can set the exact frequency to emit the TX-power, TX-antennas and the RX-antennas for signal acquisition. By clicking “Enable Test Mode”, the chip will run in CW mode with the chosen configuration.

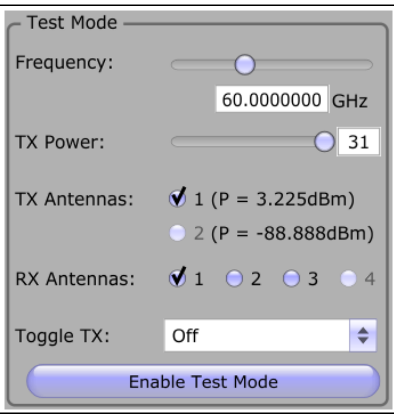

Figure 15 Test mode configuration

2.6 Other configurable parameters

As shown in Figure 2, there are a few more things to configure in the “Global” tab.

- **Trigger**: It starts the execution of a frame

- **Stop/Reset**: It stops the data acquisition
- **Data Slice Size**: The data slice size sets a fill level of the chip's FIFO memory after which an interrupt request is sent to the MCU for data polling.
- **Set and Clear Calibration**: It is possible to subtract constant (calibration) values from the samples. To do so, place the radar sensor in a calibration environment and click "Set Calibration". To clear the calibration, simply press "Clear Calibration".
- **Average over**: The GUI supports averaging of multiple frame data. Just set the value to "Average over"
- **# Frames**: Sensing GUI also enables running the frame sequence a certain number of times. Place the desired number in "# Frames". The value zero means infinite repetitions

2.7 Load and save a settings file

Once the right COM port is selected on launching Sensing GUI, a settings file can be quickly loaded into Sensing GUI and with the click of a button (Trigger button), the user is ready to detect targets. To load new settings into Sensing GUI, select "File" and "Load Settings" in the menu bar. There, select the desired file in the file browser and the settings of the configuration is loaded. To save the current settings from Sensing GUI, select "File" and "Save Settings" in the menu bar and choose a file name for the settings file.

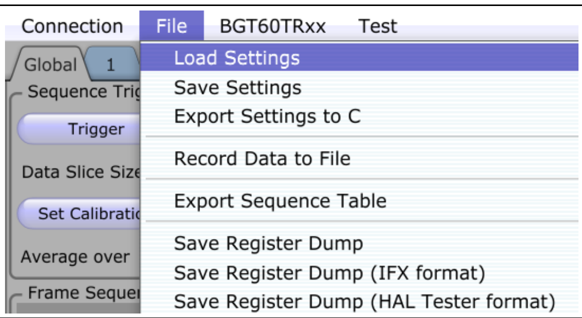

Figure 16 Loading and saving a settings file

3 References

RS, I. P. (2019). *AN599 - Radar Baseboard MCU7.* Neubiberg: Infineon Technologies.

Revision history

Document version	Date of release	Description of changes
V1.0	30/11/2019	Initial version
V1.1	28/01/2020	Update on screenshots

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